

into 40 channels, each 2MHz wide. Three of them are used for advertising (that's dedicated to connectionless communication). The other 37 are used when two connected devices communicate.

At each connection event, the used channel changes at random. So, in this way, packet transmission gets dotted around the ISM band. In addition, the system notes any non-performing channel (due to CRC failures or

spectrum interference) and removes it. Thus, efficient communication is ensured, even when a part of the ISM band is saturated. This whole behaviour is known as adaptive frequency hopping (AFH). AFH significantly reduces the probability of collisions and allows Bluetooth devices to function well even in very difficult radio environments.

As stated earlier, signal strength can greatly impact reliability and FEC is one of the techniques used to address this. But, in addition, be aware that radio receivers work best when the level of the received signal stays within a certain range (known as the golden range). There is a Bluetooth feature called LE power control, which allows connected devices to request dynamic changes to transmission power levels so that the received signal strength stays within this golden range. This is particularly useful when the distances between devices vary during normal use.

Avoiding buffer overflow

To mitigate the reliability challenges of buffer overflow, the rate of flow of packets has to be managed. This can be done using a technique known as flow control, which is used in many different communication technologies.

The Bluetooth Low Energy L2CAP layer of the stack applies flow control when required and supports a variety of different flow control modes. For example, if you are using the enhanced attribute protocol at the top of the stack, then it uses the Enhanced Credit Based Flow Control Mode within the L2CAP layer, allowing the receiving device to communicate to the transmitting device its current

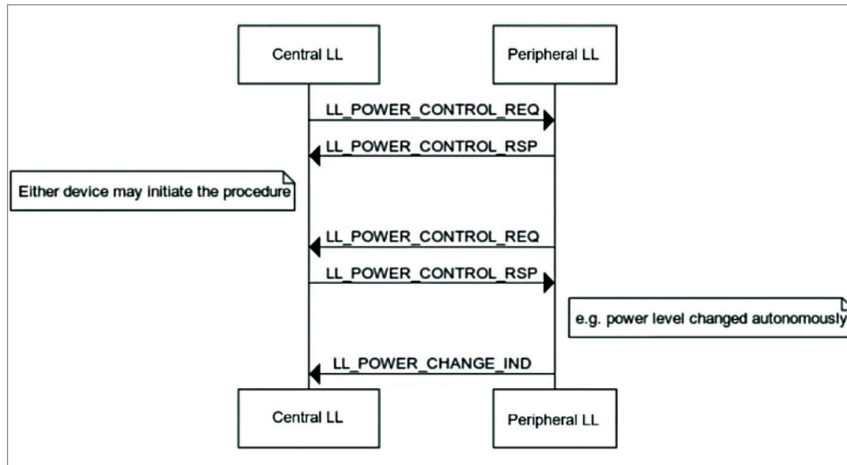


Fig. 9: Flow control

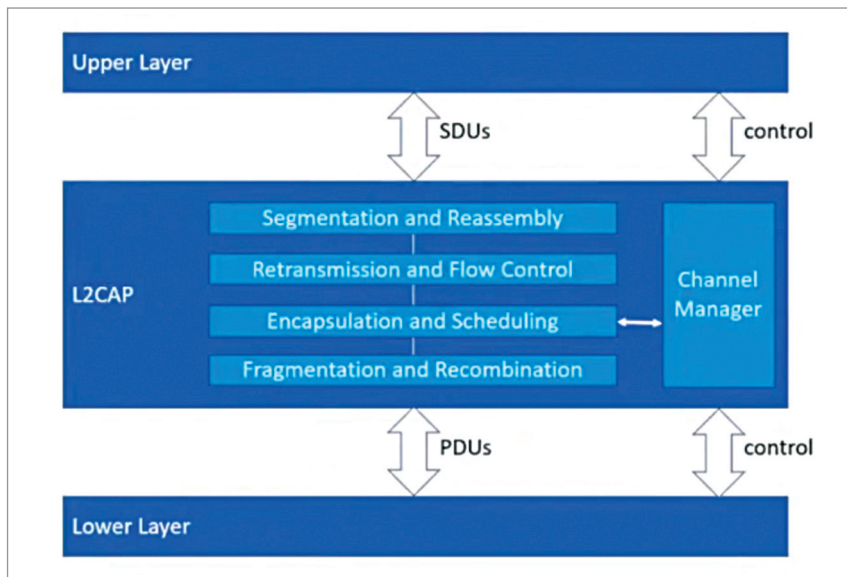


Fig. 10: Flow control technique

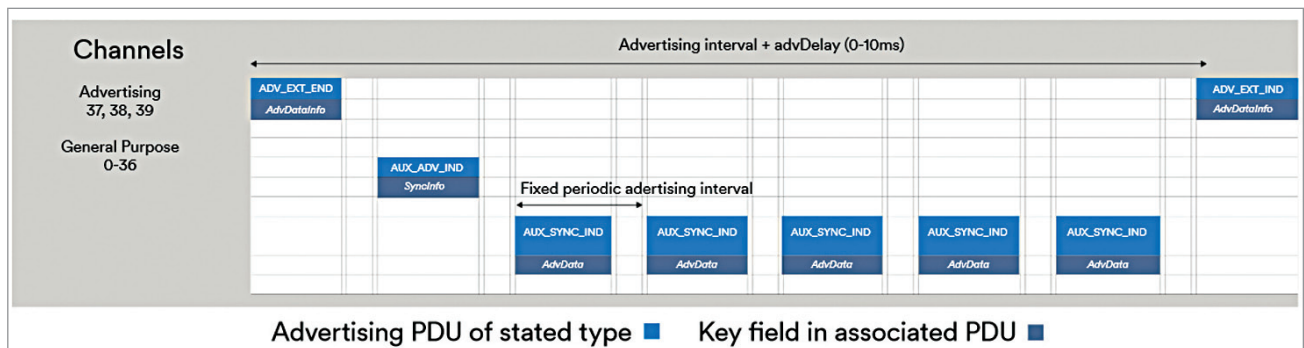


Fig. 11: Periodic advertising